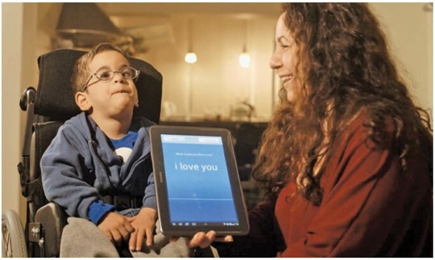

Talkitt for speech impaired (Image courtesy: cdn.nocamels.com)

at the start of a line, end of a line, moving to a new line or moving too far away from the text baseline.

Be My Eyes. This iPhone application connects the visually impaired with sighted volunteer helpers from around the world via live video connection. It is an easy way to seek help for simple tasks like checking the expiry date on a milk carton or navigating surroundings. Volunteer helper receives a notification for help and a live video connection is established. If the volunteer is too busy, the app can find someone else to step in and help.

Sesame phone. It is a touch-free smartphone designed for people with motor disabilities. It works by tracking the user's head movements, using the built-in, front-facing camera on the phone. These tracked movements are combined with computer vision algorithms to create a cursor that appears on the screen of the phone.

The on-screen cursor is controlled by the position and movement of the user's head, and supports even minimal movements. Touch, swipe, browse, play and download—it's all possible using the Sesame smartphone. Voice control adds a real hands-free experience to the phone.

Kapten PLUS personal navigation device. Traveling alone is a challenge for the visually impaired. There is always the possibility of taking a wrong turn or getting disoriented in the shuffle of busy pedestrians. The Kapten PLUS personal navigation device is a very small GPS locator designed for people with low vision. As users walk down the street, the device speaks out directions and locations, so users always know where they are and where they're heading. In addition, users can plan and store routes and tag locations for use again. Designed as an affordable GPS accessory (and not a total replacement) to cane or guide-dog travel, the device offers security, confidence and a wealth of useful information, allowing visually impaired people to travel independently without the fear of getting lost or wandering in the wrong direction.

Car with smart feedback. Engineers are developing a car that can actually be driven by the visually impaired. The aim is to integrate several computer systems, sensors and cameras to observe the environment around the vehicle and provide alternate forms of sensory input, including sound and vibration. This may include seat vibrations of various strengths and locations, pulsing vibration signals in gloves worn by the driver, auditory alerts from a headset and a sort of screen that paints a virtual picture of the surroundings using compressed air.

Smart glasses. Oxford University researchers are developing a pair of glasses that gives people with limited vision an aid that boosts their awareness of what's around them. With extra information on who or what's around them, they can walk around unfamiliar places confidently and with greater freedom. The gadget consists of two small cameras, a gyroscope, a compass, a GPS unit, a headphone and transparent OLED displays. Using it, visually impaired people would be able to distinguish between light and dark. The glasses make anything a little brighter when it gets closer to the wearers, so they can discern people and obstacles.

Devices for people with hearing impairment

Cochlear implant. This little device continues to evolve with advancements in software and hardware. The single-channel implant provided mostly static, while early commercial implants with five channels gave some indication of cadence and rhythm. Today's cochlear implants, however, have more than twenty sound channels, allowing users to hear with much better clarity. The implant is still far from perfect, with background noise continually being a problem. However, the technology has advanced to such a point now that voices can be heard with enough clarity to be readily understood and identified, making verbal communication possible and productive.

UNI. It is a two-way communication tool for hearing-impaired persons that relies upon gestures and speech technology. It works by detecting hand and finger gestures with its specialised camera algorithm, then converting these into text in a very short time to provide meaning to a given sign language. A voice recognition software converts